

MUHAMMAD TAHA BIN ZAEEM

Computer Engineering Undergraduate | NUST CEME

Applied AI Research | LLM Evaluation | Trustworthy AI | AI for Engineering | Reproducible Research
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Updated 22 August 2026 | Pakistan

RESEARCH PROFILE

Computer Engineering undergraduate focused on applied AI research that pairs language models with deterministic, testable systems. Builds research prototypes in which AI interprets unstructured requirements, proposes structured models, or explains results, while controlled experiments, native solvers/validators, and explicit evidence gates determine what can be claimed. Current work spans LLM behavior measurement, AI-assisted computer-architecture design-space exploration, AI-to-EDA workflows, and accessibility-centered learning. Turns open-ended questions into bounded datasets, reproducible pipelines, measurable experiments, and traceable research artifacts.

RESEARCH INTERESTS

Applied ML & LLMs, LLM evaluation, trustworthy AI, AI-assisted engineering/scientific workflows, human-centered AI, reproducible experimentation, computer-architecture DSE.

ACADEMIC BACKGROUND

National University of Sciences and Technology (NUST), CEME

Bachelor of Engineering in Computer Engineering | Aug 2025 - Oct 2029 (expected) | Third semester

Selected focus: computer architecture, digital logic, algorithms, programming, circuits, electronics, and machine-learning foundations.

Government College University (GCU), Lahore

Intermediate in Engineering | Grade A | Sep 2023 - Sep 2025

Qazi Grammar Boys High School

Matriculation in Computer Science | Grade A+ | Top Performer | Apr 2021 - Apr 2023

SELECTED RESEARCH & APPLIED AI EXPERIENCE

Authorial Style Separability Loss in LLM-Rewritten Fiction

Independent research project | 2026 - Present | Study and manuscript in progress; not yet published

- Designed a controlled stylometric study of how LLM rewriting changes authorial signal across lexical, syntactic, rhythmic, and distributional dimensions using six public-domain authors and 12 canonical Project Gutenberg works.
- Built a traceable data pipeline with provenance registries, checksums, conservative cleaning, sentence-aware extraction, and balanced selection of 360 passages from 1,879 candidates.
- Locked paraphrase, modernization, simplification, output-schema, and quality-control conditions before preparing 1,080 rewrite requests, reducing post-hoc drift in the experiment design.

ParetoCo - AI-Assisted Heterogeneous Computer-Architecture DSE

Co-creator | Research engineering project | 2026 - Present

- Co-built an applied research environment that combines Gecode-backed native constraint search, SDF workload modeling, and Pareto/multi-objective analysis for heterogeneous CPU/GPU/DSP/NPU/accelerator design spaces.
- Integrated Featherless.ai as a bounded research assistant for natural-language-to-structured DSE model construction and result interpretation; native and deterministic computation remains the source of truth for feasibility and trade-off evaluation.
- Implemented a reproducible experimentation layer with sessions, snapshots, semantic diffs, Pareto-front evolution, and system analyses spanning throughput, NoC, thermal, cache/memory, real-time, reliability, and diagnostics.

ProGenEDA - Hybrid AI + Deterministic EDA Generation

Founder, Product and Engineering Lead | Jul 2026 - Present

- Architected a hybrid pipeline in which an LLM interprets bounded circuit intent into a canonical CircuitIR/JSON representation, then deterministic validators and exporters compile editable native Proteus, KiCad, EasyEDA Pro, and LTspice projects.
- Built evidence-gated validation around 1,000 verified example circuits, separating structure, connectivity, geometry, native parsing, and simulation checks from claims that still require engineering review.

Controlled-Mutation Fingerprinting for Semantic Change Attribution in EDA Project Files

Independent research project | 2026 - Present | Phase 1 protocol in development; not yet published

- Designed a ground-truth-first Proteus study using save-only controls, isolated resistor mutations, and topology variants to separate file-save noise from semantic engineering edits.
- Structured a reproducible repository for extraction, file comparison, feature generation, mutation attribution, metadata, and manuscript evidence before analysis begins.

Type2Learn - Accessibility-First AI Learning Platform

Founder and Development Lead | Jul 2026 - Present

- Led development of a bilingual active-learning prototype combining learner-controlled layout, colour, motion, sound, read-aloud, and typed/spoken response paths with course-aware AI explanations, hints, simpler framings, and examples.
- Translated 13 interview scripts from nine interviewees plus structured product reviews into feature decisions while keeping adaptation consent-based, reversible, privacy-aware, and explicitly non-diagnostic.

ENGINEERING FOUNDATION

Custom 20-bit Single-Cycle Vector CPU

NUST academic team project | Verilog and Xilinx ISE | 2026

- Co-designed a Harvard-architecture processor with a custom 20-bit ISA, 16-bit scalar execution, and 64-bit SIMD operations across four 16-bit lanes; verified behavior with testbenches and waveforms.
- Owned branch-condition, data-memory, immediate-extension, lane load/store, core integration, and top-level logic; project earned Second Runner-Up in the SEMPEC Junior Hardware category.

PhishRod

Development Intern | Jul 2026 - Present

- Support software development in a security-focused product environment, strengthening production engineering discipline alongside research work.

Arch Technologies

Software Programmer Intern | Jan 2026 - Apr 2026

- Built four C++17 GUI applications for Windows and Linux using Win32 and X11, including an object-oriented ATM simulator and other interactive utilities.

RESEARCH READINESS

- Independent problem formulation and experimental design across LLM evaluation, EDA mutation analysis, and computer-architecture DSE.
- Experience building reproducible datasets and pipelines with provenance, checksums, structured schemas, deterministic validation, and explicit research boundaries.
- Comfort moving between AI/ML, systems engineering, hardware, and product contexts - useful for applied research where model behavior must be tested inside real workflows.

HONOURS AND LEADERSHIP

- Second Runner-Up, SEMPEC Junior Hardware Category, NUST CEME - custom vector CPU.
- Runner-Up, STEM 2024 - PKR 20,000 prize.
- Vice President, Dr. Samar Mubarakmand Society for Sciences, GCU Lahore | Sep 2023 - Jan 2025.
- Vice President, Dr. Khorana Society for Chemistry, GCU Lahore | 2023 - 2025.

SELECTED CREDENTIALS

- Machine Learning Specialization - DeepLearning.AI and Stanford Online | Dec 2025.
- Introduction to Logic and Critical Thinking Specialization - Duke University | Jul 2026.
- Game Theory - Stanford University and University of British Columbia | Sep 2024.

TECHNICAL SKILLS

AI / ML: Python, NumPy, Pandas, Jupyter; LLM prompting, structured outputs, and model-output evaluation.

Research: reproducible pipelines, dataset design, experimental controls, technical writing, and LaTeX.

Programming / Systems: Python, C++, JavaScript, Verilog, MIPS assembly; Git, Linux, FastAPI, and React.

Engineering: Proteus, KiCad, LTspice, Xilinx ISE, Arduino, digital logic, and circuit simulation.

RESEARCH GOAL

Develop rigorous, publishable applied-AI research that combines empirical LLM evaluation with reliable engineering systems and clear evidence about what the model can - and cannot - be trusted to do.